

12. Suppose that you wish to classify an observation $X \in \mathbb{R}$ into **apples** and **oranges**. You fit a logistic regression model and find that

$$\widehat{\Pr}(Y = \text{orange} | X = x) = \frac{\exp(\hat{\beta}_0 + \hat{\beta}_1 x)}{1 + \exp(\hat{\beta}_0 + \hat{\beta}_1 x)}.$$

Your friend fits a logistic regression model to the same data using the *softmax* formulation in (4.13), and finds that

$$\widehat{\Pr}(Y = \text{orange} | X = x) = \frac{\exp(\hat{\alpha}_{\text{orange}0} + \hat{\alpha}_{\text{orange}1}x)}{\exp(\hat{\alpha}_{\text{orange}0} + \hat{\alpha}_{\text{orange}1}x) + \exp(\hat{\alpha}_{\text{apple}0} + \hat{\alpha}_{\text{apple}1}x)}.$$

- (a) What is the log odds of **orange** versus **apple** in your model?

$$\begin{aligned} \text{(a)} \quad \text{odds} &= \frac{\widehat{\Pr}(Y = \text{orange} | X = x)}{\widehat{\Pr}(Y = \text{apple} | X = x)} = \frac{\widehat{\Pr}(Y = \text{orange} | X = x)}{1 - \widehat{\Pr}(Y = \text{orange} | X = x)} \\ \mu &= \hat{\beta}_0 + \hat{\beta}_1 x \\ &= \frac{\hat{p}(x)}{1 - \hat{p}(x)} = \frac{\frac{e^\mu}{1 + e^\mu}}{1 - \frac{e^\mu}{1 + e^\mu}} = \frac{\frac{e^\mu}{1 + e^\mu}}{\frac{1}{1 + e^\mu}} = e^\mu \\ \log \frac{\hat{p}(x)}{1 - \hat{p}(x)} &= \log e^\mu = \mu = \hat{\beta}_0 + \hat{\beta}_1 x \end{aligned}$$

- (b) What is the log odds of **orange** versus **apple** in your friend's model?

$$\begin{aligned} \text{odds} &= \frac{p}{1-p} = \frac{\frac{e^{(\hat{\alpha}_{00} + \hat{\alpha}_{01}x)}}{e^{(\hat{\alpha}_{00} + \hat{\alpha}_{01}x)} + e^{(\hat{\alpha}_{10} + \hat{\alpha}_{11}x)}}}{1 - \frac{e^{(\hat{\alpha}_{00} + \hat{\alpha}_{01}x)}}{e^{(\hat{\alpha}_{00} + \hat{\alpha}_{01}x)} + e^{(\hat{\alpha}_{10} + \hat{\alpha}_{11}x)}}} \\ &= \frac{e^{(\hat{\alpha}_{00} + \hat{\alpha}_{01}x)}}{e^{(\hat{\alpha}_{10} + \hat{\alpha}_{11}x)}} = e^{(\hat{\alpha}_{00} - \hat{\alpha}_{10}) + (\hat{\alpha}_{01} - \hat{\alpha}_{11})x} \end{aligned}$$

$$\log \frac{p}{1-p} = (\hat{\alpha}_{00} - \hat{\alpha}_{a0}) + (\hat{\alpha}_{01} - \hat{\alpha}_{a1})x$$

- (c) Suppose that in your model, $\hat{\beta}_0 = 2$ and $\hat{\beta}_1 = -1$. What are the coefficient estimates in your friend's model? Be as specific as possible.

$$\begin{array}{c} \text{2} \qquad \qquad \qquad -1 \\ \underbrace{\hspace{1.5cm}} \qquad \underbrace{\hspace{1.5cm}} \\ (\hat{\alpha}_{00} - \hat{\alpha}_{a0}) + (\hat{\alpha}_{01} - \hat{\alpha}_{a1})x = \hat{\beta}_0 + \hat{\beta}_1 x \\ \text{"} \qquad \qquad \text{"} \qquad \qquad \text{"} \qquad \qquad \text{"} \\ 2 + c_0 \qquad \qquad c_0 \qquad -1 + c_1 \qquad \qquad c_1 \end{array}$$

⇒ not only one solution

- (d) Now suppose that you and your friend fit the same two models on a different data set. This time, your friend gets the coefficient estimates $\hat{\alpha}_{\text{orange}0} = 1.2$, $\hat{\alpha}_{\text{orange}1} = -2$, $\hat{\alpha}_{\text{orange}0} = 3$, $\hat{\alpha}_{\text{orange}1} = 0.6$. What are the coefficient estimates in your model?

$$\hat{\beta}_0 = \hat{\alpha}_{00} - \hat{\alpha}_{a0} = 1.2 - 3 = -1.8$$

$$\hat{\beta}_1 = \hat{\alpha}_{01} - \hat{\alpha}_{a1} = -2 - 0.6 = -2.6$$

- (e) Finally, suppose you apply both models from (d) to a data set with 2,000 test observations. What fraction of the time do you expect the predicted class labels from your model to agree with those from your friend's model? Explain your answer.

100%. Both models make identical predictions for every test observation because in d, the softmax parameters imply the same log-odds $\beta_0 = \alpha_{00} - \alpha_{a0}$, $\beta_1 = \alpha_{01} - \alpha_{a1}$. Hence they produce the same probabilities.